Does the Distance Approach to Pairs Trading Survive in the Modern Era?

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The distance approach

- A pair trade holds offsetting positions in two assets whose prices share a long-run equilibrium, and profits when a temporary deviation closes.
- Gatev et al. (2006) match stocks with the closest normalized price paths over a formation year and trade divergences beyond 2 historical standard deviations, earning excess returns of up to 11% a year on all liquid US stocks from 1962 to 2002.
- Do and Faff (2010, 2012) found that profitability declined to 2009, mainly because a growing share of pairs never converged, with the baseline rule largely unprofitable net of costs. Matching within industries and favoring pairs with many zero crossings improved it.
- Krauss (2017) argued that ranking by the sum of squared distances selects low-variance spreads, whose trades earn little.

Research question

Question

Does the distance approach to pairs trading survive in the modern era, for US large-cap stocks at daily frequency from June 2014 to May 2024?
Does a composite score of distance, zero crossings and spread volatility improve performance?

Finding

The distance approach does not survive. No portfolio earns a mean return distinguishable from zero, the composite score and the choice among Euclidean, Manhattan and Chebyshev distances make no significant difference, and all six portfolios lose money outside the COVID half-year.

Data

Sector	Stocks	Pairs
Financial Services	24	276
Industrials	21	210
Healthcare	19	171
Consumer Cyclical	19	171
Utilities	18	153
Real Estate	17	136
Consumer Defensive	17	136
Technology	15	105
Communication Services	15	105
Basic Materials	12	66
Energy	12	66
Total	189	1,595

Stocks and within-sector pairs by sector.

- The universe is a list of 189 large-cap US stocks compiled in 2024.
- Daily adjusted closing prices run from June 2013 to May 2024.
- Pairs are formed within sectors, which gives 1,595 pairs.

Pair formation

- The sample is divided into 20 consecutive trading periods E_k of 6 months, $k = 1, \ldots, 20$, from June 2014 to May 2024, each preceded by a 12-month formation period Φ_k.
- Write $P_{i,t}$ as the adjusted closing price of stock i on day t. Within a window $W \in \{\Phi_k, E_k\}$ beginning on day τ_W, normalized prices and pair spreads are defined as

$$\tilde{P}_{i,t} = \frac{P_{i,t}}{P_{i,\tau_W}}, \qquad d_{ij,t} = \tilde{P}_{i,t} - \tilde{P}_{j,t}.$$

- A pair is eligible in period k when both stocks belong to the same sector and have a complete price record over Φ_k. Its Euclidean, Manhattan and Chebyshev distances are

$$D_{ij}^{(2)} = \Big(\sum_{t \in \Phi_k} d_{ij,t}^2 \Big)^{1/2}, \qquad D_{ij}^{(1)} = \sum_{t \in \Phi_k} |d_{ij,t}|, \qquad D_{ij}^{(\infty)} = \max_{t \in \Phi_k} |d_{ij,t}|.$$

The Euclidean distance ranks pairs as the sum of squared distances of Gatev et al. (2006).

Composite score

With T the number of formation days and σ_{ij} and $\bar{d}_{ij}$ the standard deviation and the mean of the spread over Φ_k,

$$\frac{1}{T}\sum_{t\in\Phi_k} d_{ij,t}^2 = \frac{T-1}{T}\,\sigma_{ij}^2 + \bar{d}_{ij}^2,$$

so a low distance implies a low spread volatility, as argued in Krauss (2017). With Z_{ij} the number of sign changes of $d_{ij,t}$ over Φ_k, the composite score rewards volatility and zero crossings alongside a small distance,

$$R_{ij}^{(q)} = 0.6\,\phi\big(1/D_{ij}^{(q)}\big) + 0.2\,\phi\big(Z_{ij}\big) + 0.2\,\phi\big(\sigma_{ij}\big), \qquad q \in \{2, 1, \infty\},$$

where ϕ scales a statistic linearly onto $[0, 1]$ over all eligible pairs.

Six portfolios

For each distance q and selection $m \in \{\text{distance}, \text{composite}\}$, the portfolio holds the 20 pairs with the lowest $D_{ij}^{(q)}$ or the highest $R_{ij}^{(q)}$.

Trading strategy

Prices are normalized again at the start of each trading period. The position $n_{ij,t} \in \{-1, 0, 1\}$ set at the close of day t follows the rule of Gatev et al. (2006),

$$n_{ij,t} = \begin{cases} -\operatorname{sgn}(d_{ij,t}) & \text{if } n_{ij,t-1} = 0 \text{ and } |d_{ij,t}| > 2\sigma_{ij}, \text{ and } t \text{ is not the last day of } E_k, \\ 0 & \text{if } n_{ij,t-1} \neq 0 \text{ and } d_{ij,t}\, n_{ij,t-1} \geq 0, \text{ or } t \text{ is the last day of } E_k, \\ n_{ij,t-1} & \text{otherwise.} \end{cases}$$

- A trade opened at the close of day o buys \$1 of the cheaper stock and sells \$1 of the other, then holds both until the spread crosses zero or the period ends.
- The portfolio return r_t averages the 20 pairs net of a transaction cost $c = 10$ basis points per dollar traded on each leg, so a round trip costs 40 basis points, and a borrowing fee $b = 25$ basis points a year on the short leg (Appendix B.1).

Hypotheses

Write $r_t^{(q,m)}$ for the daily return of the portfolio with distance q and selection m. We test

$$\begin{aligned}
&\text{H1:}\ \ \mathbb{E}\big[r_t^{(q,m)}\big] > 0,\\
&\text{H2:}\ \ \mathbb{E}\big[r_t^{(q,\text{composite})} - r_t^{(q,\text{distance})}\big] \neq 0,\\
&\text{H3:}\ \ \mathbb{E}\big[r_t^{(q,m)} - r_t^{(2,m)}\big] \neq 0.
\end{aligned}$$

- H1 asks whether each portfolio survives, H2 whether the composite score differs from the distance, and H3, for $q \in \{1, \infty\}$, whether the choice of distance matters.
- We test each with the Newey and West (1987) t statistic for the mean daily return or difference, one-sided for H1 and two-sided for H2 and H3.

Net performance

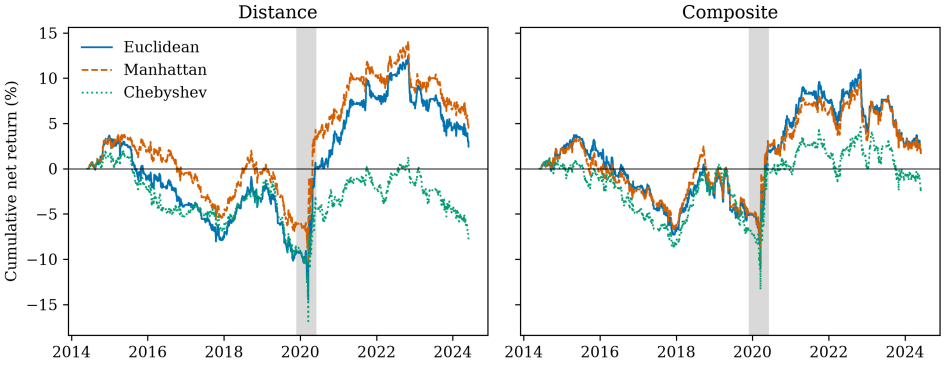
Distance	Selection	Return (%)	Volatility (%)	Sharpe	t	p
Euclidean	Distance	0.24	3.88	0.063	0.20	0.42
	Composite	0.17	3.98	0.043	0.14	0.45
Manhattan	Distance	0.44	3.91	0.11	0.36	0.36
	Composite	0.17	4.06	0.041	0.13	0.45
Chebyshev	Distance	−0.79	3.91	−0.20	−0.67	0.75
	Composite	−0.26	4.04	−0.065	−0.21	0.58

June 2014 to May 2024. Return and volatility are annualized, and p is the one-sided p value.

H1

Distance-based pair formation does not survive in the modern era. With an annualized volatility of about 4%, no portfolio earns a return distinguishable from zero.

Cumulative net return



The weakness is persistent rather than a product of the pandemic. The COVID trading period from December 2019 to May 2020, shaded, first deepens the losses as spreads widen in March 2020 and then recovers them sharply as the spreads close.

Returns before, during and after COVID

		Before COVID		COVID		After COVID		Excluding COVID	
Distance	Selection	Return	t	Return	t	Return	t	Return	t
Euclidean	Distance	−1.67	−1.41	18.9	1.07	0.58	0.33	−0.72	−0.71
	Composite	−0.92	−0.67	14.3	0.90	−0.056	−0.030	−0.56	−0.50
Manhattan	Distance	−1.10	−0.94	20.0	1.11	0.17	0.094	−0.57	−0.56
	Composite	−0.90	−0.64	15.2	0.93	−0.23	−0.12	−0.61	−0.54
Chebyshev	Distance	−1.59	−1.30	8.77	0.48	−0.85	−0.57	−1.28	−1.35
	Composite	−1.25	−0.96	13.1	0.74	−0.55	−0.32	−0.95	−0.91

Annualized net mean return (%) and t statistic from June 2014 to November 2019, in the COVID trading period from December 2019 to May 2020, from June 2020 to May 2024 and over all periods except the COVID period.

All six portfolios lose money before the pandemic and over the 19 periods without the COVID half-year, which is the best period for every portfolio although its returns are not significant. The pandemic was not the cause of the poor performance but the one episode that masked it.

Composite score and choice of distance

		Difference (%)	t	p
H2: composite minus distance selection				
Euclidean		−0.072	−0.12	0.90
Manhattan		−0.28	−0.40	0.69
Chebyshev		0.53	0.90	0.37
H3: Manhattan or Chebyshev minus Euclidean				
Manhattan	Distance	0.20	0.50	0.62
	Composite	−0.0059	−0.015	0.99
Chebyshev	Distance	−1.03	−1.60	0.11
	Composite	−0.43	−0.63	0.53

Differences in annualized net mean return, with two-sided p values.

H2 and H3

Neither refinement of the formation rule changes the conclusion, and no difference is significant. The Chebyshev distance earns about 1% a year less than the Euclidean under the distance selection, with $p = 0.11$.

Formation statistics and trades

Distance	Selection	σ (%)	Z	Trades/pair	Converged (%)	Net profit per trade (%)		
						All	Converged	Forced
Euclidean	Distance	2.55	21.5	1.23	38.6	0.10	6.32	−3.81
	Composite	2.70	25.1	1.26	40.5	0.068	6.69	−4.44
Manhattan	Distance	2.59	22.3	1.23	39.6	0.18	6.43	−3.93
	Composite	2.78	25.8	1.26	39.5	0.066	6.92	−4.41
Chebyshev	Distance	2.58	18.5	1.16	36.1	−0.34	5.80	−3.81
	Composite	2.69	23.0	1.20	37.2	−0.11	6.30	−3.91

Z is the number of zero crossings. Trades per pair counts trades per selected pair and period. A trade is converged when the spread crosses zero and forced when it is closed at the end of the trading period.

The composite score selects more volatile spreads with more zero crossings, yet opens barely more trades and closes a similar share by convergence. Only 36% to 41% of trades converge, and their gains of about 6% are offset by losses of about 4% on the rest.

Conclusion

- We tested whether distance-based pair formation survives in the modern era, forming pairs among 189 US large-cap stocks within sectors under three distances and ranking them by the distance alone or by a composite score.
- Over 20 periods of 6 months from June 2014 to May 2024, the rule of Gatev et al. (2006) does not survive. Neither the composite score nor the choice of distance improves the result, and all six portfolios lose money without the COVID half-year.
- Rewarding spread volatility enlarges the gains of converging trades and the losses of non-converging trades alike, so it does not overcome the non-convergence that erodes the strategy.

References

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Code and replication data are available at *GitHub*.

Appendix A.1. Execution with a 1-day wait

Distance	Selection	Return (%)	Sharpe	t
Euclidean	Distance	0.073	0.019	0.060
	Composite	−0.062	−0.016	−0.051
Manhattan	Distance	0.36	0.097	0.30
	Composite	−0.13	−0.032	−0.10
Chebyshev	Distance	−1.02	−0.27	−0.91
	Composite	−0.50	−0.13	−0.42

Net performance with execution 1 day after the signal, June 2014 to May 2024.

- Executing a trade at the close that generates its signal may overstate returns through bid-ask bounce, which Gatev et al. (2006) address by waiting 1 day. This table repeats the analysis with every position change executed at the close of the following day.
- Every portfolio earns less than in the net performance table, no mean is distinguishable from zero and the composite score again makes no significant difference, with $|t|$ at most 0.93 for H2.

Appendix B.1. Net portfolio return

A trade opened at the close of day o buys \$1 of the cheaper stock and sells \$1 of the other. The portfolio return is the average over its 20 pairs,

$$r_t = \frac{1}{20} \sum_{(i,j)} \left[n_{ij,t-1} \left(\frac{P_{i,t} - P_{i,t-1}}{P_{i,o}} - \frac{P_{j,t} - P_{j,t-1}}{P_{j,o}} \right) - 2c \left| n_{ij,t} - n_{ij,t-1} \right| - \frac{b}{252} \frac{P_{s,t-1}}{P_{s,o}} \left| n_{ij,t-1} \right| \right],$$

- s is the stock held short.
- The transaction cost is $c = 10$ basis points per dollar traded on each leg, so a round trip costs 40 basis points.
- The borrowing fee is $b = 25$ basis points a year on the value of the short leg.